

Objective

Seeking an entry-level role in embedded systems programming, firmware engineering, or software engineering.

Available May 2026 • Raleigh, NC • Open to relocation

Focus: low-level C/C++, microcontroller bring-up, RTOS fundamentals

Technical Skills

Languages: C, C++, Python, Bash, Verilog, VBA, MATLAB, LC-3 Assembly

Embedded/Tools: Code Composer Studio, PlatformIO, Git, KiCad, WaveForms, SSH, JTAG, Beagle Analyzer

Buses/Peripherals: CAN, UART/RS232, SPI, I²C, ADC, PWM, Timers

Systems: FreeRTOS, Embedded Linux, Arch Linux, WSL2, UEFI/BIOS

Hardware: Soldering, PCB/debug, oscilloscopes, logic/protocol analyzers

Professional Experience

Aegis Power Systems – Computer Science Intern

May–Aug 2024 & May–Aug 2025

- Designed an SSH-controlled CLI; integrated RS232 and CAN; contributed to FreeRTOS-based firmware.
- Built a remote firmware update path: file generation, checksum/packetized transfer, and SPI reflashing.
- Created a TI TMS320 serial flasher (CLI) with usage docs; repaired legacy BDC code for internal use.
- Standardized Git practice across teams: GUI-first workflow (GitLab + Git Extensions), GitFlow-style branches, and signed annotated tags; authored an 80+ page illustrated guide + handout + slides.
- Developed PWM/ADC control in CCS: 12 PWM outputs with a config manager and ADC-driven duty cycle.

TEAM Industries – Engineering Intern

Jun–Aug 2022

- Automated QC spreadsheets in VBA to flag out-of-spec parts; maintained spec sheets and documentation.

Embedded & Firmware Projects

Embedded Systems Toy Car (ECE 306) – MSP430, C, CCS, custom PCB

- Built and programmed a toy car using a TI MSP430 microcontroller and custom PCBs.
- Integrated IoT Wi-Fi control, ADC-based line following, PWM motor control, and button/switch inputs with software debouncing.
- Programmed an LCD to display essential system information in real time.

Vending Machine Logic Circuit (ECE 212) – Sequential logic, WaveForms/Analog Discovery 2

- Designed and built a sequential logic circuit on a breadboard.
- Verified behavior on a WaveForms test bench using an Analog Discovery 2.
- Reduced gate count and simplified state transitions for reliability.

Variable Output Boost Converter (555 Timer)

- Prototyped and characterized with an oscilloscope; tuned frequency/duty cycle for target output.

Software Projects

C++ Maze Solvers (ECE 309) – BFS, DFS, GBFS

- Implemented and benchmarked search algorithms; compared time/memory across maze topologies.

Quantum Search (ECE 469) – Qiskit on WSL Arch

- Set up a reproducible Qiskit/Aer environment on WSL Arch; documented install and troubleshooting.
- Implemented Grover's algorithm; analyzed success vs. iteration count against a classical baseline.

Leadership & Communication

OPS 2 Lead, IEEE @ NCSU

Fall 2025

- Coordinated hands-on prototyping topics (KiCad, soldering, BOMs, custom PCBs, IoT/low-power).
- Planned projects and distribution logistics; contributed to officer events.

Peer Support & Mentoring (ECE 306)

- Assisted classmates with MCU bring-up, driver debugging, and CCS workflows; shared reproducible setups.

Math Tutor, Tri-County Community College

- Taught pre-calculus, statistics, and quantitative literacy; adapted to each student's needs.
- Dedicated 2–3 hours per day supporting remote and in-person learners.

Education

North Carolina State University, Raleigh, NC — *B.S. Computer Engineering*

May 2026

Selected: Embedded Systems, Digital Logic, C/C++, Microprocessor Architecture, Operating Systems

Tri-County Community College, Murphy, NC — *Associate of Arts*

Fall 2024